

Project:
**Advancing Integrated Materials and Devices for Electrochemical
CO₂ Reduction (AIM-COR)**

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July 2026 report

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July monthly report

1. To-do list in July

- 100 cm² MEA electrolyzer operation in 15° tilted position – CO₂ electrolysis of 0.5NiSAC (850 °C-4h-10 °C·min⁻¹) catalyst in AEM system with 350 μm / 250 μm silicone gaskets and 10 ft·lb under 50 mM CH₃COOH humidified 400 mL/min CO₂, 100 mL/min of 0.1 M KHCO₃, and IrO₂-TiPt anode at 100 mA/cm² — long-term durability
- Exploring the possible high current density operation in 5cm² AEM electrolyzer: CO₂ electrolysis at current density range of 100 – 300 mA/cm² and long-term stability test

2. Experimental Procedure

Synthesis of 0.5NiSAC by pyrolysis method

The synthesis of 0.5NiSAC catalyst was followed by our previous work. Initially, nickel phthalocyanine (NiPc), urea, and dicyanamide precursors were stoichiometrically weighed according to mass ratios of 0.5:5:5. Subsequently, the mixture was thoroughly mixed and ground until a homogeneous midnight blue powder was obtained. The fine powder was then transferred into a quartz boat covered with a capsule lid and subjected to polymerization at 800 - 900 °C (ramp rate = 10 °C·min⁻¹) in a horizontal tube furnace under an argon atmosphere (flow rate > 200 sccm) 2 - 4 h.

Preparation of AgNPs cathode via spray coating method

The NiSAC gas-diffusion electrode was fabricated by spray coating onto Sigracet 39BB carbon paper or hydrophilic carbon cloth substrates. Typically, 150 mg of AgNPs and 381 μL of XC-2 ionomer were ultrasonically dispersed in 10 mL of isopropanol (IPA) for at least 1 h to form a homogeneous dark colloidal suspension. The ink suspension was subsequently spray-coated onto a 12 x 12 cm² carbon substrate using an automated, custom-built spraying setup equipped with a syringe pump operating at a flow rate of 0.6 mL min⁻¹ on a hotplate maintained at 130°C, yielding a catalyst mass loading of 0.5 ± 0.1 mg.cm⁻² AgNPs.

Electrochemical measurements in membrane-electrode assembly (MEA) electrolyzer

5 and 100cm² MEA electrolyzer operational set-up

A 100 cm² MEA electrolyzer was used to evaluate the CO₂RR performance under modulated operating conditions. The IrO₂-TiPt felt (10.5 x 10.5 cm²), AEM (12 x 12 cm²), and AgNPs (10.5 x 10.5 cm²) were employed as the anode, membrane, and cathode electrode, respectively. The MEA was assembled with 0.25μm PTFE gasket and tightened at a compression force of 8 ft.lb. 10 L of 0.1 M KHCO₃ was recirculated and used throughout the experiment as the

analyte at 100 ml/min. The humidified CO₂ gas stream was introduced into the cathodic inlet with a total CO₂ flow rate of 200-800 ml/min. Prior the test, the electrochemical impedance spectroscopy (EIS) and Cyclic voltammetry were measured. Meanwhile, the cathodic outlet was directly injected into an online gas-chromatography for gas products analysis.

3. Results

Effect of MEA tilt angle on long-term FE(CO) durability at 100 cm²

To evaluate the effect of MEA tilt angle on the long-term operational durability of the 0.5NiSAC catalyst, the 100 cm² cell was operated galvanostatically at 100 mA·cm⁻² (50 mM CH₃COOH humidified CO₂, 100 mL·min⁻¹ of 0.1 M KHCO₃) in a 0° (vertical) and a 15° tilted configuration. As shown in Figure 1, tilting the MEA cell from 0° to 15° markedly extended the stable FE(CO) operation from ~20 h to ~590 h, with FE(CO) maintained at ~95–100 % and the cell voltage held at ~3.0–3.2 V throughout the tilted run, while the single-pass CO₂ utilisation stabilised at ~18–20 %. This pronounced improvement is attributed to the more effective gravity-assisted drainage of liquid water achieved under 15° tilted operation, which mitigates water accumulation within the gas-diffusion electrode (GDE) — one of the main causes of rapid cell failure in the vertical configuration. Near ~590 h, a gradual rise in cell voltage toward ~4 V was observed, accompanied by an increase in the CA-outlet gas signal and in parasitic HER activity, indicating the eventual onset of GDE flooding as accumulated water progressively restored the HER pathway at the expense of CO₂ reduction. Post-mortem inspection of the disassembled cell (Figure 2) revealed no precipitated salt within the flow-field channels, confirming that carbonate/bicarbonate salt precipitation was not the primary cause of cell failure in this case and corroborating water-driven GDE flooding as the limiting degradation mechanism. Overall, the ~590 h of stable ~95–100 % FE(CO) represents the longest continuous 100 cm² operation achieved to date and demonstrates that a modest 15° tilt substantially enhances the water management and durability of the 0.5NiSAC/AEM system.

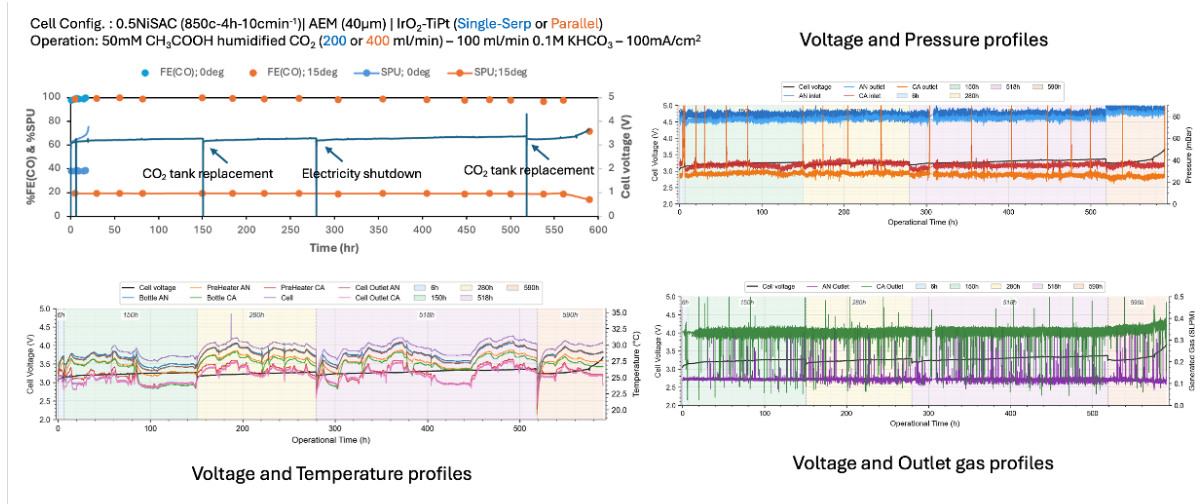


Figure 1. Long-term CO₂ electrolysis of the 0.5NiSAC (850 °C, 4 h, 10 °C·min⁻¹) / AEM 40 μm / IrO₂-TiPt 100 cm² MEA at 15° tilt, showing FE(CO) and SPU (0° single-serpentine, 200 mL·min⁻¹ vs 15° parallel, 400 mL·min⁻¹), cell voltage, CA-outlet pressure, cell temperature, and outlet-gas concentration versus operational time under 50 mM CH₃COOH humidified CO₂, 100 mL·min⁻¹ of 0.1 M KHCO₃, at 100 mA·cm⁻².

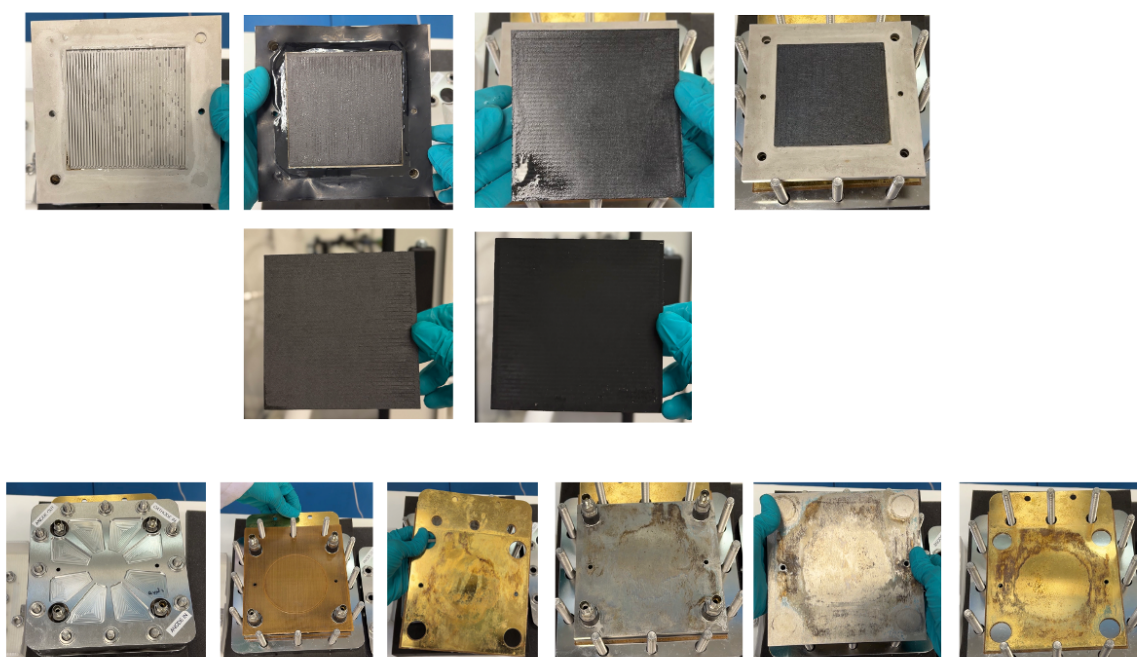


Figure 2. Post-mortem disassembly of the 100 cm² MEA after 600 h of CO₂ electrolysis, showing the cathode flow-field plate, the 0.5NiSAC gas-diffusion electrode (front and back), the AEM membrane, and the anode flow-field plate and housing.

Exploring the possible high current density operation in 5cm² AEM electrolyzer

The 5 cm² MEA cell (0.5NiSAC cathode, IrO₂-TiPt anode) was revisited and operated at high current density range of 100–300 mA·cm⁻² for 1 h at each current density on a Sustainion

AEM (0.5 M CH_3COOH humidified CO_2 at $20 \text{ mL} \cdot \text{min}^{-1}$, $4 \text{ mL} \cdot \text{min}^{-1}$ of 0.1 M KHCO_3). As shown in Figure 3, $\text{FE}(\text{CO})$ remained above 95 % from 100 to $250 \text{ mA} \cdot \text{cm}^{-2}$, with the single-pass CO_2 utilisation rising to a maximum of 46.14 % at $250 \text{ mA} \cdot \text{cm}^{-2}$ (cell potential 2.94–3.28 V). Increasing the current density further to $300 \text{ mA} \cdot \text{cm}^{-2}$ lowered $\text{FE}(\text{CO})$ to 86.94 % while the SPU increased only marginally to 48.75 %, indicating that the reaction had entered a mass-transport-limited regime under the high-SPU, CO_2 -depleted condition.

To evaluate the consequences of operating in this depleted- CO_2 zone, long-term stability tests were subsequently conducted on a PiperION AEM ($40 \mu\text{m}$). At $300 \text{ mA} \cdot \text{cm}^{-2}$, $\text{FE}(\text{CO})$ declined rapidly from ~ 85 % to ~ 65 % within only ~ 7 h of operation, consistent with the dominance of the competing HER. At $250 \text{ mA} \cdot \text{cm}^{-2}$, the cell sustained $\text{FE}(\text{CO}) > 90$ % for ~ 12 h before (bi)carbonate salt clogging of the flow-field channels caused cell failure; notably, raising the humidifier concentration from 0.5 M to 1.0 M CH_3COOH did not prevent the salt-induced failure. These results demonstrate that operating in the high-SPU, CO_2 -depleted regime accelerates cell failure through the combined effects of massive (bi)carbonate salt precipitation and the consequent GDE flooding. Accordingly, further operational conditions — increasing the acetic acid humidifier concentration or adopting a stronger-acid humidifier, together with increasing the CO_2 flow rate — will be explored to relieve CO_2 depletion and extend the operational lifespan of the system.

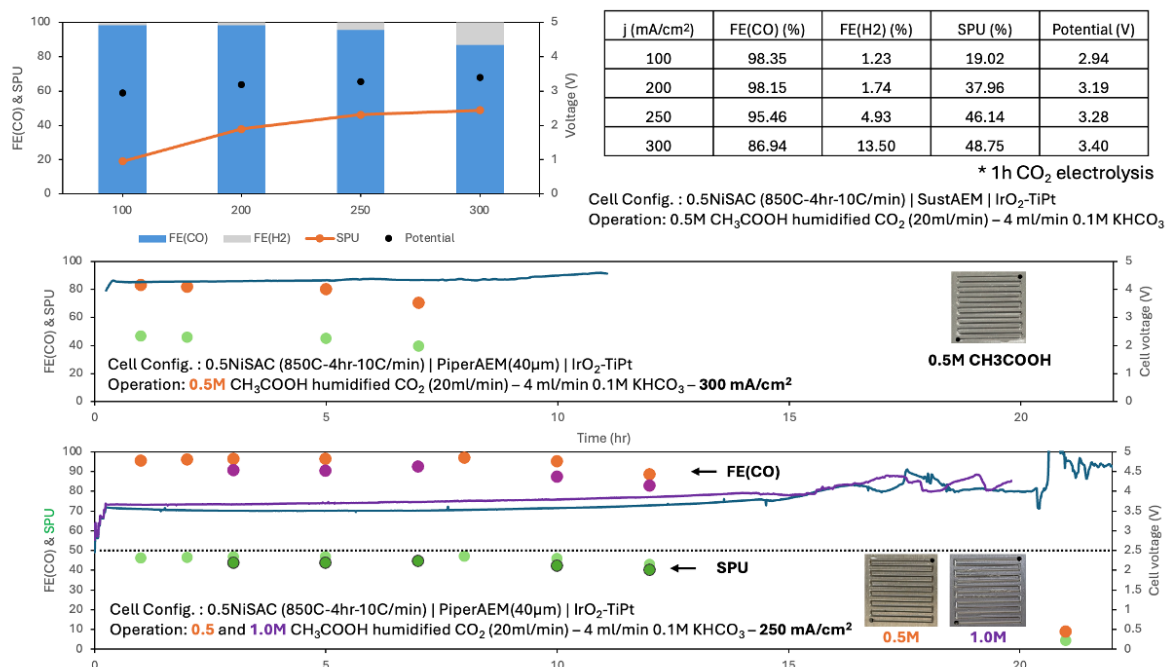


Figure 3. CO_2 electrolysis of 0.5NiSAC (850°C , 4 h, $10^\circ\text{C} \cdot \text{min}^{-1}$) / IrO_2 -TiPt in a 5 cm^2 MEA at different applied current densities: (top) $\text{FE}(\text{CO})$, $\text{FE}(\text{H}_2)$, SPU, and cell potential after 1 h electrolysis at 100, 200, 250, and $300 \text{ mA} \cdot \text{cm}^{-2}$ on a Sustanion AEM; (middle) long-term $\text{FE}(\text{CO})$, SPU, and cell voltage at $300 \text{ mA} \cdot \text{cm}^{-2}$ and (bottom) at $250 \text{ mA} \cdot \text{cm}^{-2}$ comparing 0.5

M and 1.0 M CH_3COOH humidifier, both on a PiperION 40 μm AEM. Conditions: 0.5 M CH_3COOH humidified CO_2 at $20\text{ mL}\cdot\text{min}^{-1}$ (1.0 M for the bottom-panel comparison), $4\text{ mL}\cdot\text{min}^{-1}$ of 0.1 M KHCO_3 .

Optimization of Ni-SAC catalyst

Effect of pyrolysis temperature and hold time on the surface area, CO_2 -to-CO selectivity, and durability of 0.5NiSAC

To optimise the 0.5NiSAC synthesis, the effect of pyrolysis temperature (800, 850, and 900 $^\circ\text{C}$) and hold time (2 and 4 h) on the textural properties and CO_2RR performance was investigated at a fixed ramp rate of $10\text{ }^\circ\text{C}\cdot\text{min}^{-1}$. As shown in Figure 4, the Brunauer–Emmett–Teller (BET) analysis revealed that the pyrolysis temperature directly governed the textural properties: raising the temperature from 800 to 900 $^\circ\text{C}$ progressively increased both the specific surface area ($135.70 \rightarrow 171.50 \rightarrow 181.92\text{ m}^2\cdot\text{g}^{-1}$ for the 2 h series and $112.69 \rightarrow 178.11 \rightarrow 210.36\text{ m}^2\cdot\text{g}^{-1}$ for the 4 h series) and the pore volume ($0.34 \rightarrow 0.41 \rightarrow 0.52\text{ cm}^3\cdot\text{g}^{-1}$ and $0.36 \rightarrow 0.53 \rightarrow 0.61\text{ cm}^3\cdot\text{g}^{-1}$, respectively). In contrast, extending the hold time from 2 to 4 h exerted only a modest effect on the specific surface area while substantially increasing the pore volume (for example, from 0.41 to $0.53\text{ cm}^3\cdot\text{g}^{-1}$ at 850 $^\circ\text{C}$). In terms of CO_2RR performance, the 0.5NiSAC synthesised at 850 $^\circ\text{C}$ for 4 h (850C-4h- $10\text{ }^\circ\text{C}\cdot\text{min}^{-1}$) outperformed all other conditions, delivering the highest FE(CO) and the longest operational lifetime. These results indicate that an intermediate pyrolysis temperature combined with the extended 4 h hold time provides the optimal balance of active-site accessibility and mass-transport-favourable porosity for durable CO_2 -to-CO conversion.

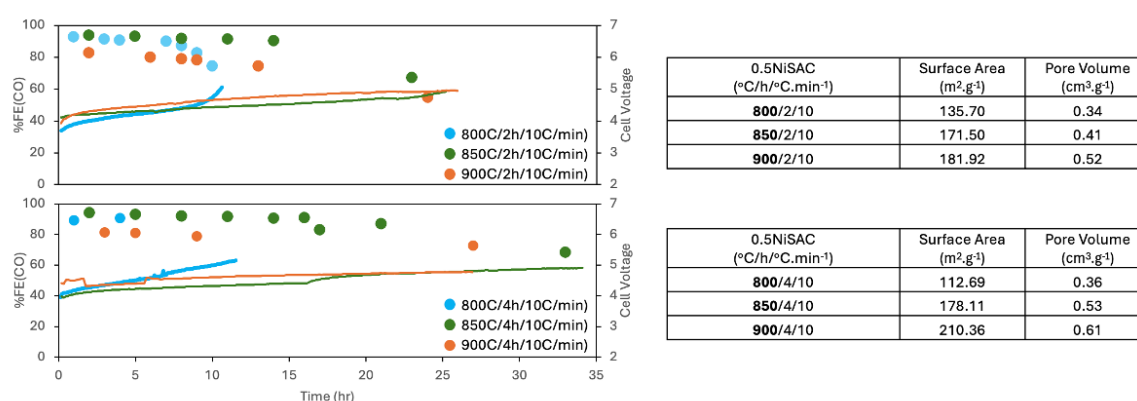


Figure 4. Optimisation of 0.5NiSAC synthesis — effect of pyrolysis temperature and hold time. FE(CO) and cell voltage versus operational time for 0.5NiSAC pyrolysed at 800, 850, and 900 $^\circ\text{C}$ for (top) 2 h and (bottom) 4 h at a fixed $10\text{ }^\circ\text{C}\cdot\text{min}^{-1}$ ramp rate, with the corresponding BET specific surface area and pore volume tabulated.

Effect of pyrolysis ramp rate on the surface area, CO₂-to-CO selectivity, and durability of 0.5NiSAC

To further probe the synthesis parameters, the effect of the heating ramp rate was examined at each pyrolysis temperature, as shown in Figure 5. The BET analysis showed no consistent trend in the textural properties with ramp rate across the temperatures tested: at 850 °C (2 h), increasing the ramp rate from 10 to 20 °C·min⁻¹ raised the specific surface area from 171.50 to 220.90 m²·g⁻¹ with only a marginal change in pore volume (0.41 to 0.43 cm³·g⁻¹), whereas at 900 °C (4 h), increasing the ramp rate from 5 to 10 °C·min⁻¹ slightly lowered both the specific surface area (216.87 to 210.36 m²·g⁻¹) and the pore volume (0.79 to 0.61 cm³·g⁻¹). Consistently, the CO₂RR performance exhibited no systematic dependence on ramp rate, with FE(CO) and durability varying inconsistently between conditions. Because the present two-point ramp-rate comparison at each temperature is insufficient to resolve a clear trend, the ramp-rate range will be extended (5, 10, and 20 °C·min⁻¹) to more rigorously establish its effect on the catalyst texture and CO₂-to-CO performance.

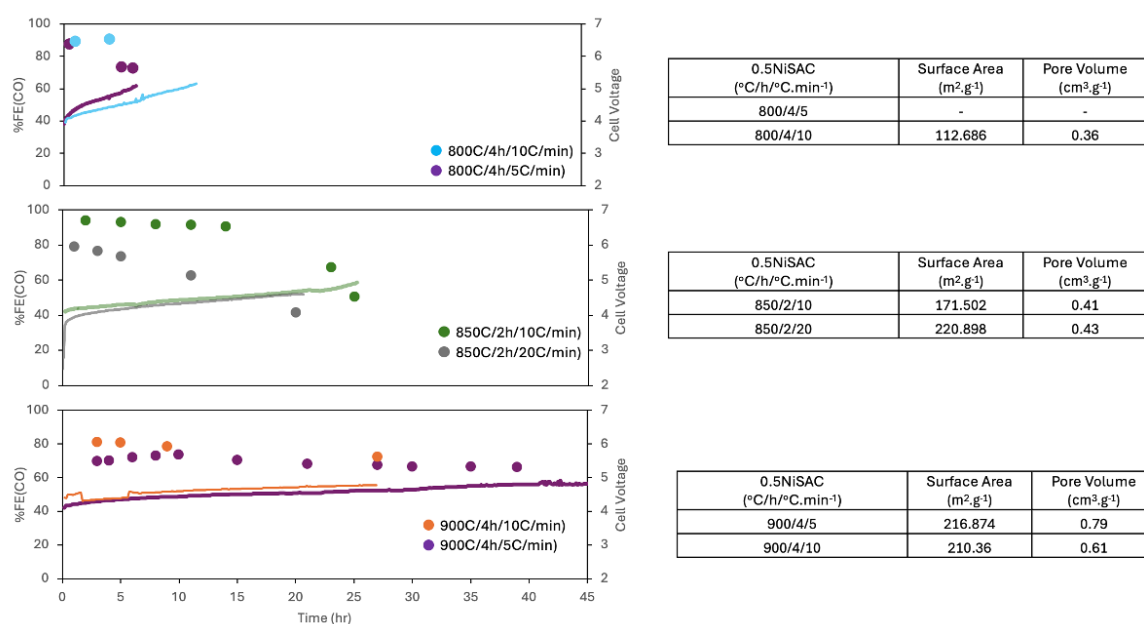


Figure 5. Optimisation of 0.5NiSAC synthesis — effect of pyrolysis ramp rate at each temperature. FE(CO) and cell voltage versus operational time for 0.5NiSAC pyrolysed at (top) 800 °C / 4 h (5 vs 10 °C·min⁻¹), (middle) 850 °C / 2 h (10 vs 20 °C·min⁻¹), and (bottom) 900 °C / 4 h (5 vs 10 °C·min⁻¹), with the corresponding BET specific surface area and pore volume tabulated.

Upcoming tasks

- **100cm² MEA electrolyzer operation in 15 position** – CO₂ electrolysis of 0.5NiSAC (850C-4h-10 C.min⁻¹) catalyst in AEM system with 350μm/250μm silicone gaskets and 8 ft.lbs under 50mM CH₃COOH humidified 400 ml/min CO₂, 100 ml/min of 0.1M KHCO₃, and IrO₂-TiPt anode at **200mA/cm²**
- **5cm² MEA electrolyzer. – CO₂ electrolysis of 0.5NiSAC** (850C-4h-10 C.min⁻¹) in AEM system at 250mA/cm²: Exploring the possible operational conditions (Effect of CO₂ flow rate and acetic acid concentration)